The Power of Prior Revisited: Training-Free Open-Vocabulary Segmentation with LLaVA

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Abstract

Training-free open-vocabulary segmentation asks a frozen vision-language model (VLM) to provide semantic and spatial evidence that can be converted into pixel masks without segmentation-specific fine-tuning. We independently reimplement and evaluate the FSeg-LLaVA framework, which extracts class-conditioned spatial priors from LLaVA and uses them to prompt the frozen Segment Anything Model (SAM). On all 1,449 PASCAL VOC 2012 validation images, our Hugging Face implementation achieves 65.85 VOC21 mIoU, close to the paper's 68.0 and to 66.01 from the released implementation. The method transfers across LLaVA versions: LLaVA-1.6 Vicuna and Mistral obtain 65.02 and 66.29 mIoU, respectively. On 5,000 COCO-Object images, the same pipeline obtains 39.99 mIoU. Oracle class recognition raises performance to 69.46, 70.00, and 50.41 on the two VOC model variants and COCO, respectively. These gaps show that class-presence recognition, rather than mask refinement alone, is the dominant bottleneck. Layer-wise analysis further shows that middle-layer maps are compact and precise seeds for SAM, whereas deeper maps have better standalone coverage but less useful prompts. Our contribution is a controlled reproduction and cross-model, cross-dataset analysis that identifies where the training-free prior succeeds and where it fails.

1 Introduction

Open-vocabulary semantic segmentation requires a system to assign pixels to classes specified at inference time. Conventional segmentation systems achieve strong accuracy by training on pixel-level annotations, but this makes their class vocabulary and deployment cost restrictive. Recent vision-language models (VLMs) contain broad semantic knowledge and image-text alignment, while promptable segmenters such as SAM provide high-quality masks from sparse spatial prompts. The FSeg-LLaVA framework combines these capabilities without training a segmentation head: LLaVA answers whether a candidate class is present and its internal representations provide a coarse spatial prior; SAM turns that prior into pixel masks.

This work studies whether the proposed mechanism is reproducible and whether its spatial prior generalizes beyond the exact model and dataset used in the original paper. We ask:

1. Can the published result be reproduced in an independent Transformers implementation?
2. Which implementation details and LLaVA layers determine the quality of SAM prompts?
3. Does the method transfer across LLaVA variants and from VOC to COCO-Object?
4. Is final performance limited primarily by recognition or by mask generation?

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Our study makes four empirical contributions. First, we provide a controlled reproduction on the complete VOC validation split and compare against the released implementation. Second, we separate recognition, localization, and SAM refinement through coarse-map and oracle-class experiments. Third, we perform a layer-wise study that distinguishes seed precision from standalone map coverage. Fourth, we evaluate multiple LLaVA variants and a larger, more diverse COCO-Object benchmark. We deliberately present this as a reproduction-plus-analysis study, not as a claim to have invented the original segmentation algorithm.

2 Literature Review

Vision-language models. CLIP-style contrastive pretraining aligns image and text representations, while instruction-tuned models such as LLaVA connect a vision encoder to a large language model for visual question answering. These models are trained for semantic understanding rather than pixel-level prediction, but their image-token representations can retain spatial information.

Open-vocabulary segmentation. Open-vocabulary and referring segmentation methods typically transfer image-text alignment to dense prediction, often using additional training, region proposals, or specialized decoders. Training-free approaches instead exploit frozen representations and prompting, making them useful for studying the latent spatial capabilities of general-purpose VLMs.

Promptable segmentation. SAM is a general-purpose promptable segmenter trained on large-scale mask data. It does not decide which object class is present; it refines points, boxes, and masks into object boundaries. The FSeg-LLaVA pipeline therefore separates semantic recognition and localization from boundary generation.

Position of this work. The original FSeg-LLaVA paper demonstrates the core pipeline. We extend its evaluation with an independent implementation, explicit paper-versus-code ablations, oracle recognition controls, layer diagnostics, cross-version model comparisons, and COCO-Object transfer. This makes the failure modes and generalization limits measurable rather than attributing all errors to the segmentation stage.

3 Problem Statement

Given an image I and a candidate vocabulary $\mathcal{C}$, the goal is to predict an indexed mask $\hat{Y} : I \to \{0, \ldots, |\mathcal{C}|\}$ without training on segmentation labels for the target vocabulary. The system may use frozen pretrained models, but the inference procedure must determine which classes are present and where their pixels occur.

We evaluate with mean intersection-over-union:

$$mIoU = \frac{1}{|\mathcal{C}| + 1} \sum_{c=0}^{|\mathcal{C}|} \frac{|\hat{Y} = c \cap Y = c|}{|\hat{Y} = c \cup Y = c|},$$

where class 0 is background and void pixels are ignored. All primary VOC numbers use VOC21 scoring (20 foreground classes plus background), with intersections and unions accumulated over the complete split.

4 Proposed Method

4.1 Question–answer pipeline

For each candidate class c, LLaVA receives a fixed image question asking whether c is present. Positive answers are retained and provide the class token and a short description. Negative answers are discarded. This stage supplies a semantic gate, but it can introduce false positives and false negatives before any mask is generated.

Table 1: VOC21 validation results. "Coarse" evaluates the LLaVA spatial prior without SAM. Oracle recognition supplies the correct present-class set only for diagnosis.

System	Coarse	Standard	Oracle recognition
Paper-reported LLaVA-1.5	–	68.00	–
Our LLaVA-1.5 reproduction	18.72	65.85	72.16
Released implementation	–	66.01	–
LLaVA-1.6 Vicuna	16.45	65.02	69.46
LLaVA-1.6 Mistral	19.90	66.29	70.00

4.2 Text–visual response

Let h_c^l be the hidden state of the class-name token at language-model layer l, and let h_i^l be the hidden state of image token i. We compute class-conditioned spatial scores from cosine similarities:

$$D_c(i) = \frac{1}{|L|} \sum_{l \in L} \frac{h_c^l \cdot h_i^l}{\|h_c^l\| \|h_i^l\|}.$$

The scores are reshaped to the vision-token grid and thresholded. We additionally use description-to-image attention to retain spatially coherent evidence. Our default follows the released implementation where the attention map is combined with the thresholded map by union, rather than the product described in the paper text.

4.3 Visual prompting and SAM

CLIP patch features are used to form class and background prototypes. The resulting coarse mask provides positive points, a box, and negative points associated with other classes. SAM ViT-H refines these prompts for three rounds. Candidate masks are selected using overlap with the class-conditioned prior and then merged using an explicit background policy.

4.4 Independent implementation

Our implementation uses Hugging Face Transformers for LLaVA access, explicit token indexing, cached per-image traces, and a separate evaluation module. Hyperparameters are selected on a 200-image subset of VOC train and frozen before full validation. Oracle experiments replace only the predicted present-class set with the ground-truth set and are reported as diagnosis, never as the primary result.

5 Experiments and Results

5.1 Experimental setup

The primary VOC experiments use LLaVA-1.5-7B with SAM ViT-H and all 1,449 VOC2012 validation images. We also evaluate LLaVA-1.6 Vicuna and Mistral on the same split. The COCO-Object run contains 5,000 validation images and uses the same training-free pipeline. Experiments run on a single NVIDIA A30 GPU. No segmentation-specific fine-tuning is performed.

5.2 Reproduction and model comparison

Table 1 shows that the independent reproduction is within 2.15 points of the paper and within 0.16 points of the released implementation. The two LLaVA-1.6 models remain close to the LLaVA-1.5 reproduction, indicating that the approach is not restricted to one checkpoint. Mistral is the strongest tested standard model at 66.29 mIoU.

5.3 COCO-Object transfer

The full pipeline transfers to COCO with 39.99 mIoU despite the larger vocabulary, smaller objects, and more crowded scenes. SAM improves the coarse prior by 21.59 points. Correct class recog-

Table 2: COCO-Object validation results on 5,000 images.

Configuration	mIoU
LLaVA coarse maps only	18.40
Full LLaVA + SAM pipeline	39.99
Oracle recognition + SAM	50.41

Table 3: Selected VOC2012 ablations from the independent LLaVA-1.5 reproduction.

Configuration	VOC21 mIoU
Full reproduction	65.85
No prototype step	65.78
No attention mask	65.49
Paper-text map product	61.62
SAM points only	57.70
SAM box only	54.12
Class token shifted one position	48.57
Coarse LLaVA maps only	18.72

nition would add another 10.42 points, demonstrating that recognition quality becomes even more important as the vocabulary and visual diversity increase.

5.4 Ablations and implementation checks

The ablations reproduce the qualitative trends in the original work. The union used by the released code is substantially better than the product written in the paper text. Reading the class token at the wrong position costs 17.28 points, showing that the spatial prior is sensitive to exact token alignment. SAM is essential: the coarse map is not a competitive final mask, but it is a useful prompt for a general-purpose segmenter.

5.5 Layer-wise analysis

The layer study separates two properties that are conflated by a single final score. Layers 7–13 produce compact, precise seeds and are effective for SAM prompting. Deeper layers, especially around 17–22, produce maps with greater standalone coverage but oversized or less selective prompts. Thus, the layer with the best dense map is not necessarily the layer that gives the best downstream segmentation. This supports the interpretation that LLaVA's useful spatial prior is distributed and changes from precise localization to broad semantic coverage across depth.

5.6 Failure analysis

On VOC, the largest errors are person close-ups and recognition mistakes for chair, sofa, dining table, and bottle. In close-ups, LLaVA may localize only a face or skin region, causing SAM to refine an incomplete prior. The oracle experiments show that correcting recognition alone yields 4.43 points for LLaVA-1.6 Vicuna, 3.71 points for Mistral, and 10.42 points on COCO. This is evidence that future work should improve the semantic gate and class-conditioned prior, rather than treating all residual error as a SAM boundary problem.

6 Discussion

The results support three conclusions. First, the core method is reproducible: both our independent implementation and the released implementation are close to the reported result. Second, the method generalizes across LLaVA variants and from VOC to COCO, although absolute performance decreases on the more difficult COCO setting. Third, the system is bottlenecked by semantic recognition and spatial-prior quality. SAM is highly effective once it receives useful prompts, but it cannot recover an object class that was rejected or a prior that covers only a small part of an object.

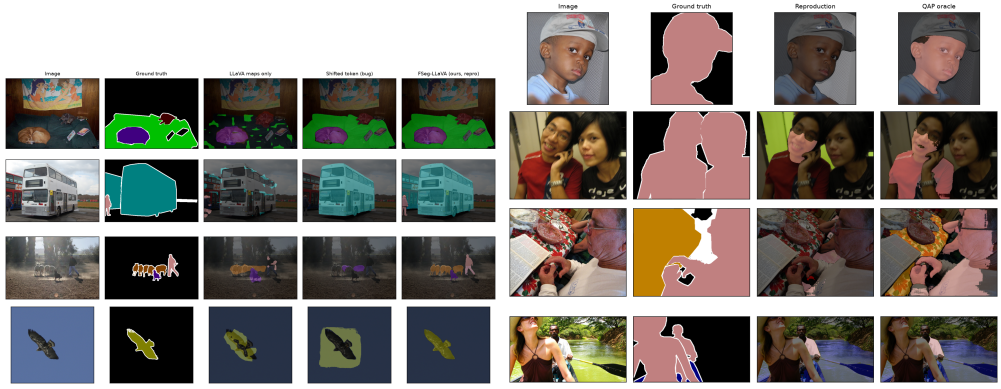


Figure 1: Representative qualitative success and failure cases from the VOC evaluation. The failure cases illustrate incomplete priors and recognition errors.

The main scientific value of this study is therefore diagnostic. It establishes a practical decomposition of training-free VLM segmentation into recognition, localization, and boundary refinement. This decomposition avoids the misleading conclusion that a low coarse-map score means that the VLM contains no useful spatial information: the COCO and oracle results show that weak priors can still produce substantially better masks when coupled to SAM.

7 Limitations

Our study does not claim state-of-the-art segmentation performance. The COCO evaluation is limited to the configured COCO-Object split and should not be compared with methods trained specifically for full COCO panoptic or semantic segmentation without matching protocols. We evaluate a small number of VLM families, and the original paper's checkpoint and preprocessing environment are not identical to every modern implementation. Oracle recognition is a diagnostic upper-bound analysis, not a deployable method. Finally, tracing thousands of images with a 7B VLM is computationally expensive, which limits exhaustive prompt and layer searches.

8 Conclusions

We independently reproduced the training-free FSeg-LLaVA framework and evaluated it across VOC, COCO-Object, and multiple LLaVA versions. The main VOC reproduction reaches 65.85 VOC21 mIoU, close to both the reported 68.0 and the released implementation's 66.01. LLaVA-1.6 Vicuna and Mistral remain competitive, while COCO-Object confirms transfer beyond VOC. The central result is the recognition-versus-segmentation decomposition: SAM converts weak LLaVA spatial priors into useful masks, but recognition errors and incomplete priors remain the dominant limitations. Layer-wise measurements show that precise middle-layer seeds can be more valuable than deeper layers with better standalone coverage. Together, these findings support training-free VLM segmentation as a viable research direction and provide concrete targets for future work: improve class-presence recognition, calibrate spatial priors, and select layers according to downstream prompt quality rather than map quality alone.

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